Lec-10

21/03/26 9:47 AM

Search → Linear → one by one

[Binary search] → Sorted element

8:00am 8:30 8:38 8:45 9:00am

item = 11

[2, 3, 4, 7, 8, 9, 11, 13, 14, 15, 19, 21]

lo = 0 hi

```
while (lo <= hi) {
    int mid = (lo+hi)/2
    if (arr(mid) == item) {
        return mid
    }
    else if (arr(mid) > item) {
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

9<12 11>12 14>12 13>12

```
x = 1
while (x^k <= n) {
    x++
}
return (x-1)
```

x = 2 3 4 5 6

k = 3

N = 149

$1^3 \le 149$

$2^3 \le 149$

$3^3 \le 149$

$4^3 \le 149$

$5^3 \le 149$

$6^3 \le 149$

=> x = 5

f(x) => $\boxed{x^k \le N}$

n = 168

k = 3

K = ✓

N = ✓

x → maximum

ans: mid → 5

1 4 5 8 17 37 49 75 149

```
while (lo <= hi) {
    int mid = (lo+hi)/2
    if (mid^k <= N) {
        ans = mid
        lo = mid+1
    }
    else {
        hi = mid-1
    }
}
```

$75^3 \le 149$ ×

$37^3 \le 149$

$4^3 \le 149$

$8^3 \le 149$

N = 100

bad = 30

N = 100

```
for (i = 1; i <= n; i++) {
    if (isbadversion(i)) {
        return i
    }
}
```

bad = 30

1 25 28 29 31 36 37 49 50 100

```
while (lo <= hi) {
    mid = (lo+hi)/2
    if (isbadversion(mid)) {
        ans = mid
        hi = mid-1
    }
    else {
        lo = mid+1
    }
}
```

Constraints:

- `1 <= bad <= n <= 2^31 - 1`

$N = 2^{31}-1$

$bad = 2^{31}-3$

lo = 1 $hi = 2^{31}-1$

$\frac{x}{2} = y$

$x = 2 \times y$

$\frac{hi}{2} + \frac{lo}{2} = \frac{7}{2} + \frac{3}{2}$

$\frac{(2^{31}-1) + (2^{31}-7)}{2}$

$= \frac{2^{31}-1 + 2^{31}-7}{2} = \frac{2 \times 2^{31} - 8}{2}$

$2^{31}-4$

$mid = \frac{lo+hi}{2}$

$= \frac{(hi-lo)}{2} + lo = \frac{hi - lo + 2lo}{2} = \frac{hi+lo}{2}$

3

[1, 2, 8, 4, 9] → [1, 2, 4, 8, 9]

1 2 4 8 9

min = 1: c_1 c_2 c_3

min = 2: c_1 × c_2 c_3

min = 3: c_1 c_2 c_3

min = 4: c_1 × × c_2

```
min = 1
while (isitpossible(stall, cow, min)) {
    min++
}
```

min-1

[1, 2, 4, 8, 9]

ans = 3

lo = 1 hi = 8

→ last - first

arr(n-1) - arr(0)

```
while (lo <= hi) {
    mid = (lo+hi)/2
    if (isitpossible(stall, cow, mid)) {
        ans = mid
        lo = mid+1
    }
    else {
        hi = mid-1
    }
}
```

mid = 2

cow = 1 2 3

pos = stall(0)

```
for (i = 1; i < stall.len; i++) {
    if (stall(i) - pos >= mid) {
        cow++
        pos = stall(i)
    }
    if (cow == noc) {
        return true
    }
}
```

fun

arr(mid) > x

arr(mid) < arr(hi)

arr(mid) > item

arr(hi) item